

VPA IB DP CS — HL Extension Materials

HL Extension Materials HL扩展材料 HL Kuòzhǎn Cáiliào

For: Higher Level (HL) Students | Topics: 5, 6, 7 + Cross-Curricular Enrichment

Introduction

HL students study three additional topics (5, 6, 7) and sit Paper 3 (Case Study). This document provides extension reading, advanced problems, research prompts, and enrichment resources to support deeper understanding and potential exploration for the Extended Essay or university preparation.

1. Topic 5: Abstract Data Structures — Extension Materials

Advanced Reading

"Introduction to Algorithms" (CLRS) — Chapters 10 (Elementary Data Structures), 12 (Binary Search Trees), 22 (Graph Algorithms)

"Algorithms" by Robert Sedgewick & Kevin Wayne — Excellent for visual learners; includes Java implementations

VisuAlgo (visualgo.net) — Interactive visualisations of all data structures and algorithms

Extension Problems

Problem	Skill	Difficulty
Implement a doubly linked list with sentinel nodes to eliminate edge cases	Linked list mastery	Medium
Implement an AVL tree with automatic rebalancing after insertion	Tree balancing	Hard
Solve the "Travelling Salesman Problem" using brute force and then compare with greedy approximation	Graph algorithms, complexity	Hard
Implement Dijkstra's algorithm using a priority queue (min-heap)	Advanced graph algorithms	Hard
Design a graph to model VPA's campus and find the shortest path between any two buildings	Real-world application	Medium

Research Prompts

Why do different programming languages implement data structures differently? Compare Python lists, Java ArrayLists, and C++ vectors.

How does a database index use B-trees (a variant of binary search trees) to achieve fast lookups?

What is the "memory hierarchy" effect on data structure choice? When is an array faster than a linked list in practice?

2. Topic 6: Resource Management — Extension Materials

Advanced Reading

"Operating Systems: Three Easy Pieces" (Remzi Arpaci-Dusseau) — Free online; excellent for scheduling and memory management

"Modern Operating Systems" (Tanenbaum) — Chapters on process management and memory

OSDev Wiki (wiki.osdev.org) — For students interested in building their own OS

Extension Problems

Problem	Skill	Difficulty
Calculate turnaround and waiting times for 5 processes using FCFS, SJF, Round Robin (quantum=4), and Priority. Which is "fairest"?	Scheduling analysis	Medium
Given a 32-bit system with 4KB pages, calculate the page table size. How does it change with 64-bit?	Memory calculations	Medium
Simulate FIFO, LRU, and Optimal page replacement on a reference string. Compare page fault rates.	Virtual memory	Medium
Design a semaphore solution for the "Dining Philosophers" problem	Concurrency classic	Hard
Model a smart home thermostat as a control system with feedback loop	Control systems	Medium

Research Prompts

How does Linux's Completely Fair Scheduler (CFS) differ from traditional Round Robin?

What are the security implications of memory paging (e.g., Meltdown and Spectre vulnerabilities)?

How do real-time operating systems (RTOS) guarantee deadlines in automotive or aerospace applications?

3. Topic 7: Control — Extension Materials

Advanced Reading

"Distributed Systems" (Tanenbaum & Van Steen) — Chapters on architectures, communication, and consistency

"Designing Data-Intensive Applications" (Martin Kleppmann) — Excellent for distributed systems and CAP theorem

"Life 3.0" (Max Tegmark) — AI, autonomous systems, and future implications

Extension Problems

Problem	Skill	Difficulty
Design a consensus algorithm for 3 nodes deciding on a shared value. What happens if one node fails?	Distributed consensus	Hard

Problem	Skill	Difficulty
Compare CAP theorem trade-offs for a banking app vs a social media feed	System design	Medium
Analyse Tesla's Autopilot as an autonomous agent using the sense-plan-act model	Autonomy analysis	Medium
Debate: Should lethal autonomous weapons be banned? Prepare arguments for both sides	Ethics	Medium
Design a distributed system for VPA's attendance tracking across multiple campuses	Real-world application	Hard

Research Prompts

How does Bitcoin's Proof of Work achieve distributed consensus without a central authority?

What is the "alignment problem" in AI, and why do researchers consider it an existential risk?

How do swarms of drones coordinate without a central controller? What emergent behaviours appear?

4. Paper 3 Case Study — Extension Resources

IB Case Study (current year): Released by IB in September prior to exams

Harvard Business Review — Technology and strategy cases

MIT Sloan Management Review — Digital transformation articles

Wired, MIT Technology Review — Current technology trends

Journal of ACM / IEEE Computer — For deep technical background (optional)

5. Programming Enrichment

Advanced Languages to Explore

LanguageWhy ExploreResource

CMemory management, pointers, OS-level programming"C Programming Language" (K&R)

JavaOOP, enterprise systems, IB-friendly pseudocode similarityOracle Java Tutorials

HaskellFunctional programming, recursion, type systems"Learn You a Haskell for Great Good"

PrologLogic programming, declarative paradigms"Learn Prolog Now" (free online)

Competitive Programming

LeetCode — Algorithmic problem solving (start with "Easy", progress to "Medium")

Codeforces — Competitive programming contests

USACO Guide — Structured competitive programming curriculum

Project Euler — Mathematical/computational problems

6. University Preparation

MOCs: CS50 (Harvard), MIT 6.0001 (Python), Stanford Algorithms Specialization

Mathematics: Ensure strong calculus and discrete mathematics preparation

Reading: "The Pattern on the Stone" (Daniel Hillis) — elegant introduction to CS concepts

Research: Consider an Extended Essay in Computer Science; start exploring topics in Year 1

7. HL Student Checklist

- ☐ I have completed all extension problems for Topic 5
- ☐ I have completed all extension problems for Topic 6
- ☐ I have completed all extension problems for Topic 7
- ☐ I have read at least one book from the advanced reading list
- ☐ I have explored one new programming language beyond Python
- ☐ I have solved at least 20 LeetCode problems
- ☐ I have researched my Paper 3 case study using at least 3 external sources
- ☐ I have considered a topic for my Extended Essay
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